NeuroSym-BD: Neuro-Symbolic Violence Analysis from Live News Portals

Decoupled Description Logic Constraint Gating for Auditable Conflict Monitoring

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85.3% BinAcc | **96.8% Precision@HV** | **1.0% Std-FPR** | **100% W3C PROV-O**

The AI Dilemma in Automated Conflict Monitoring

Unconstrained Neural Sequence Models

- Flexible for journalistic prose and dialectal syntax.
- **Catastrophic False Alarms**: Over-sensitized to militant political rhetoric and hyperbole.
- *"Politicians vow a bloodbath in the upcoming elections"* misclassified as armed mass battle (*FP*).
- Zero-Shot NLI False Positive Rate: **31.3%**.

Pure Symbolic Rule Engines

- High recall on literal keyword matching (**90.2%**).
- **Severe Over-Triggering**: Incapable of parsing complex sentence syntax, passive voice, or irony.
- Triggers violent categories on non-violent administrative sentences.
- False Positive Rate: **86.9%** (Precision: 34.8%).

The NeuroSym-BD Solution

A decoupled Type 1/2 neuro-symbolic architecture: dense neural representations gated by formal Description Logic ($\mathcal{SROIQ}$) plausibility priors to slash false alarms while preserving active conflict continuity.

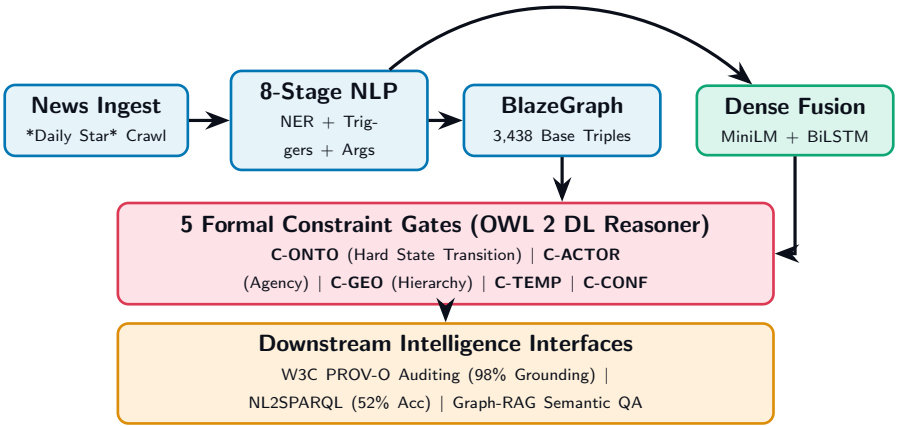
Extending CAMEO for Bangladesh Conflict Realities

Standard CAMEO (Codes 13–20) was engineered for interstate diplomacy, omitting regional South Asian violent conflict archetypes. We formulated **21 specialized OWL 2 subclasses** across **6 macro-categories**:

Macro-Category	Prefix / Subcodes	Bangladesh Socio-Political Context
Political Violence (PV)	`cameo-bd:PV01--04`	Hartal strike violence, AL vs BNP clashes, election violence.
State Violence (SV)	`cameo-bd:SV01--04`	Extrajudicial killings (*Crossfire*), custodial torture, protest crackdown.
Communal Violence (CV)	`cameo-bd:CV01--03`	Religious minority temple attacks, land grabbing disputes, CHT unrest.
Gender Violence (GV)	`cameo-bd:GV01--04`	Acid violence, domestic fatalities, sexual assault, dowry abuse.
Labor Violence (LV)	`cameo-bd:LV01--03`	Garment (RMG) wage unrest, industrial highway occupations, union suppression.
Organized Crime (OC)	`cameo-bd:OC01--03`	Mob lynching (*Gonopituni*), extortion racket clashes, drug syndicates.

Formally verified via `dl_reasoner_check.py` with **0 consistency violations**.

End-to-End System Pipeline



OWL 2 DL State Transition Plausibility Priors

Formal Description Logic ($\mathcal{SROIQ}$) Axioms

$$\exists \text{hasCategory}.\top \sqsubseteq \text{Event}, \quad \top \sqsubseteq \forall \text{hasCategory}.\text{CameoCategory}$$

$$\text{ValidEscalation} \equiv \text{HighViolence} \sqcap \exists \text{precededBy}.\text{CoercivePrecurso}$$

$$\text{UnescalatedJump} \equiv \text{HighViolence} \sqcap \exists \text{precededBy}.\text{Protest}$$

$$\text{UnescalatedJump} \sqsubseteq \text{OntologicalViolation}$$

Role disjointness:
$\text{DisjointRoles}(\text{validTransitionTo}, \text{illegalTransitionTo})$.

8×8 Transition Plausibility Matrix

From \ To	Coerce	Assault	Fight	MassV
Threaten (13)	✓	×	×	×
Protest (14)	✓	×	×	×
Force (15)	✓	✓	×	×
Coerce (17)	✓	✓	✓	×
Assault (18)	✓	✓	✓	✓
Fight (19)	×	✓	✓	✓
MassV (20)	×	×	✓	✓

Mechanism: Bayesian district transition prior preventing single-clause ungrounded escalations.

Constrained Decoding Fallback:

$$c^* = \arg \max_{c \in \mathcal{C}_{\text{valid}}(\text{prev})} P_{\text{neural}}(c)$$

Controlled Architectural Ablation ($N = 150$ Test Events)

Architecture	BinAcc	Prec@HV	Rec@HV	F1@HV	Std-FPR	Provenance
Keyword Heuristic	0.453	0.363	0.804	0.500	0.727	100.0%
Pure Symbolic (Rule Engine)	0.393	0.348	**0.902**	0.503	0.869	100.0%
Zero-Shot NLI (DistilBERT)	0.620	0.446	0.490	0.467	0.313	0.0%
Zero-Shot NLI + C-ONTO	**0.780**	**0.737**	0.549	**0.629**	**0.101**	100.0%
Dense Neural (MiniLM+LSTM)	0.787	0.852	0.451	0.590	0.040	0.0%
NeuroSym-BD Hybrid (Ours)	**0.853**	**0.968**	0.588	**0.732**	**0.010**	**100.0%**

- **Statistical Significance**: Paired binary correctness McNemar test yields $b = 12$ wins vs $c = 2$ losses ($p = 0.0129 < 0.05$).
- **75% False Alarm Reduction**: Cuts standard false positive rate from 4.0% $\rightarrow$ **1.0%** ($FP : 4 \rightarrow 1$).
- **Model-Agnostic Utility**: C-ONTO gating lifts off-the-shelf Zero-Shot NLI F1 by **+16.2 points** without retraining.

Generalization: Leakage-Resistant Article Partition

Split Methodology (Zero Article Overlap)

- Disjoint split across publisher articles:
  - 49 Training Articles ($n = 136$ events)
  - 49 Held-Out Test Articles ($n = 164$ events)
- Guarantees evaluation is not inflated by shared journalistic style, author syntax, or district reporting clusters within articles.

Metric	Neural-Only	NeuroSym-BD
Binary Accuracy	81.7%	**84.1%**
Precision@HV	89.3%	**93.3%**
Recall@HV	48.1%	**53.8%**
F1@HV	0.625	**0.683**
Std-FPR	1.8%	**1.2%**

Paired discordances: $b = 6$ wins vs $c = 2$ losses ($p = 0.289$, exact binomial test). Confirms precision gain generalizes across unseen articles.

Fine-Grained 8-Class Root Evaluation & Confusion Matrices

Reconciled Confusion Matrix (Hybrid, $n = 150$)

Act \ Pred	13	14	15	16	17	18	19	20	Tot
13 (Threat)	0	2	0	4	11	0	0	0	17
14 (Prot)	0	6	0	10	32	1	0	0	49
15 (Force)	0	0	0	1	1	0	0	0	2
16 (Reduce)	0	2	0	3	9	0	0	0	14
17 (Coerce)	0	4	0	5	8	0	0	0	17
18 (Assault)	0	0	0	0	5	14	2	0	21
19 (Fight)	0	0	0	0	4	2	2	0	8
20 (MassV)	0	1	0	0	11	2	8	0	22
Total Pred	0	15	0	23	81	19	12	0	150

Per-Category Performance Highlights

- **Violent Assault (CAMEO 18)**:
  - Precision: 54.2% → **73.7%** (**+19.5% abs**)
  - F1 Score: 0.578 → **0.700** (**+12.2% abs**)
- **Coerce (CAMEO 17)**:
  - Recall: 41.2% → **47.1%**
- **Operational Trade-off**:
  - Overall exact-match accuracy: 24.0% → 22.0% (33/150).
  - Macro-Averaged F1: 0.174 → **0.177**.

NeuroSym-BD is specifically optimized for high-stakes violence screening and false-alarm suppression rather than fine-grained nominal non-violent subcoding.

Explainability & Provenance Auditing Rubric

4-Tier Auditing Breakdown ($N = 100$ Sample)

Auditing Property	**Count**	**Ra**
Metadata Completeness (PROV-O)	98/100	98.0%
Verbatim Sentence Grounding	98/100	98.0%
Symbolic Rule Trace Coverage	100/100	**100.**
General Conflict Lexicon	98/98	100.
Category Trigger Faithfulness	85/98	**86.7**

Provenance Grounding Formulation

$$\text{Score}(e) = \text{Conf}(e) \cdot \mathbf{1}_{\text{prov}}(e) \cdot \left[\mathbf{1}_{\text{faithful}}(e) + 0.70 \cdot (1 - \mathbf{1}_{\text{faithful}}(e)) \right]$$

Mean Grounding Score: **0.721** across evaluated events.

Auditing Transparency

In 13 cases (13.3%), sentences referenced judicial context while active violence was inferred via district sequence continuity. The PROV-O audit flags these so analysts can verify evidence.

Real-World Qualitative Walkthroughs

Case 1: University Campus Quota Clashes (Dhaka)

"Over a dozen student activists were injured yesterday when clashes broke out between rival factions..."

- Neural scorer predicts CAMEO 182 (*Physical Assault*).
- Preceding district context: active police blockade and administrative standoff (prev_code = 17, Coerce).
- ValidEscalation $\equiv$ HighViolenceEvent $\sqcap \exists$precededBy.CoercivePrecursor satisfied $\rightarrow$ certified as admissible.

Case 3: Labor Unrest vs. Armed Assault Disambiguation (Savar)

"Hundreds of readymade garment workers blocked the Dhaka-Aricha highway demanding unpaid wages."

- Neural text scorer over-reacts to tense rhetoric, predicting CAMEO 182 (*Physical Assault*, *FP*).
- Preceding district event was peaceful labor demonstration (prev_code = 14, Protest).
- Abrupt jump 14 $\rightarrow$ 18 violates transition prior cameo:illegalTransitionTo.
- **C-ONTO Interception**: Falls back via constrained decoding to CAMEO 144 (*Block passage*, `cameo-bd:LV01`), successfully suppressing false alarm.

External Corroboration & Downstream Interfaces

External Conflict Benchmarks

- **ACLED (2020–2026 Archive)**:
  - 42 spatio-temporally matched events.
  - **78.6%** Exact CAMEO Code Concordance.
  - **100.0%** Macro-Family Concordance ($HV \geq 18$).
- **GDELT 2.0 (Identical Source URLs)**:
  - 78 identical story URLs evaluated without code filtering.
  - **61.5%** Exact CAMEO Root Concordance.
  - **100.0%** Macro-Family Concordance ($HV \geq 18$).

Downstream Operations

- **NL2SPARQL Benchmark ($N = 50$ Gold Queries)**:
  - 100.0% Syntax Validity.
  - 100.0% Schema Compliance.
  - **52.0%** Exact Answer Set Accuracy.
- **Knowledge Graph Embeddings (PyKEEN)**:
  - RotatE achieves **MRR = 0.369** and **Hits@10 = 0.641** (outperforming TransE 0.211).

Conclusion & Key Takeaways

Summary of Contributions

1. **Localized Ontology**: 21 BD-CAMEO subclasses grounded in South Asian socio-political realities.
2. **False Alarm Suppression**: Cuts standard FPR by 75% (4.0% $\rightarrow$ 1.0%) with 96.8% precision on high violence.
3. **Auditable Provenance**: 98% verbatim sentence grounding with full W3C PROV-O auditability.
4. **Operational Verification**: Validated on GDELT, ACLED, and NL2SPARQL.

Target Venues & Code Release

- **Target Venues**: CASE Workshop @ ACL/EMNLP; ESWC Resources Track.
- **Open Source Repository**:
  - Complete pipeline, ontologies, and test suite.
  - **28/28 Unit Tests Passing** (`pytest`).
- `https://github.com/ShMazumder/neuro-symbolic-ai-workspace.git`

Thank You! Questions?